INCH-POUND
MIL-I-24768/26
25 February 1992

# MILITARY SPECIFICATION SHEET

## INSULATION, PLASTIC, LAMINATED, THERMOSETTING, PAPER-BASE, EPOXY-RESIN (PEE)

*This specification is approved for use by all departments and agencies of the Department of Defense.*

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

**REQUIREMENTS:**

**Construction:** The material shall consist of a paper-base filler impregnated and bonded with an epoxy-resin compound or binder and processed to conform to this specification.

**Intended use:** The laminated material covered by this specification is a moisture and humidity resistant plastic laminate suitable for cold punching and for mechanical support in electrical applications. It possesses high dielectric strength and relatively low dielectric losses under conditions of adverse humidity and temperature.

**Forms:** Sheets are the only recommended form of laminated material covered by this specification.

**Burning rate:** The burning rate for the material covered by this specification shall be class 1.

**Temperature indices:** The laminated material covered by this specification has a mechanical strength test index of 90 and an electrical strength test index of 90 for thicknesses up to 0.062 inch, and a mechanical strength test index of 110 and an electrical strength test index of 110 for thicknesses greater than 0.062 inch.

**Dimensions and part numbers:**

*Thickness of sheets:* The thickness of laminated sheets, permissible variations, and the applicable part number shall be as specified in table I.

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TABLE I. *Thickness of laminated sheets.*

| Part number | Nominal thickness (inches) | Permissible variations (± inch) |
|---|---|---|
| M24768/26-S-1 | 0.010 | 0.002 |
| M24768/26-S-2 | 0.015 | 0.0025 |
| M24768/26-S-3 | 0.020 | 0.003 |
| M24768/26-S-4 | 0.025 | 0.0035 |
| M24768/26-S-5 | 0.031 | 0.0035 |
| M24768/26-S-6 | 0.047 | 0.0045 |
| M24768/26-S-7 | 0.062 | 0.005 |
| M24768/26-S-8 | 0.094 | 0.007 |
| M24768/26-S-9 | 0.125 | 0.008 |
| M24768/26-S-10 | 0.156 | 0.009 |
| M24768/26-S-11 | 0.188 | 0.010 |
| M24768/26-S-12 | 0.219 | 0.011 |
| M24768/26-S-13 | 0.250 | 0.012 |
| M24768/26-S-14 | 0.312 | 0.014 |
| M24768/26-S-15 | 0.375 | 0.017 |
| M24768/26-S-16 | 0.438 | 0.019 |
| M24768/26-S-17 | 0.500 | 0.021 |
| M24768/26-S-18 | 0.625 | 0.024 |
| M24768/26-S-19 | 0.750 | 0.027 |
| M24768/26-S-20 | 0.875 | 0.030 |
| M24768/26-S-21 | 1.000 | 0.033 |
| M24768/26-S-22 | 1.125 | 0.035 |
| M24768/26-S-23 | 1.250 | 0.037 |
| M24768/26-S-24 | 1.375 | 0.039 |
| M24768/26-S-25 | 1.500 | 0.041 |
| M24768/26-S-26 | 1.625 | 0.043 |
| M24768/26-S-27 | 1.750 | 0.045 |
| M24768/26-S-28 | 1.875 | 0.047 |
| M24768/26-S-29 | 2.000 | 0.049 |
| M24768/26-S-30 | 2.125 | 0.052 |
| M24768/26-S-31 | 2.500 | 0.056 |
| M24768/26-S-32 | 2.750 | 0.061 |
| M24768/26-S-33 | 3.000 | 0.065 |
| M24768/26-S-34 | 3.500 | 0.073 |
| M24768/26-S-35 | 4.000 | 0.082 |
| M24768/26-S-36 | 4.500 | 0.090 |
| M24768/26-S-37 | 5.000 | 0.095 |
| M24768/26-S-38 | 5.500 | 0.105 |
| M24768/26-S-39 | 6.000 | 0.115 |
| M24768/26-S-40 | 6.500 | 0.120 |
| M24768/26-S-41 | 7.000 | 0.130 |
| M24768/26-S-42 | 7.500 | 0.140 |
| M24768/26-S-43 | 8.000 | 0.145 |
| M24768/26-S-44 | 8.500 | 0.155 |
| M24768/26-S-45 | 9.000 | 0.160 |
| M24768/26-S-46 | 9.500 | 0.170 |
| M24768/26-S-47 | 10.000 | 0.180 |
| M24768/26-S-48 | _[1] | _[2] |

See footnotes on next page.



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[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.8125 inch thick sheet is: M24768/26-S-48-0.8125.
[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

**Performance requirements:** Performance requirements for laminated materials shall be as specified in table II.

The following performance requirements do not apply to this plastic laminate:

a. Surface resistance
b. Volume resistivity
c. Silicone content
d. Arc resistance
e. Thermal endurance
f. Flammability
g. Tracking resistance
h. Impact strength
i. Bonding strength.

**Testing:** The laminated material covered by this specification shall be tested under first article inspection requirements.



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TABLE II. *Performance requirements for sheets.*

| Requirement | Cond. | Test para. | Unit | Thickness (inches)[1] | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 0.031 | 0.062 | 0.094 | 0.125 | 0.188 | 0.250 | 0.500 | 0.750 | 1.000 | 1.001 – Max. |
| Dielectric breakdown voltage | A<br>D-48/50 | 4.6.4 | Min.<br>kV | 60.0<br>30.0 | 60.0<br>30.0 | 60.0<br>30.0 | 60.0<br>30.0 | 60.0<br>30.0 | 60.0<br>30.0 | 60.0<br>30.0 | 60.0<br>30.0 | 60.0<br>30.0 | |
| Dielectric constant | A<br>D-24/23<br>D-48/50 | 4.6.6 | Max. | 4.6<br>4.8 | 4.6<br>4.8 | 4.6<br>4.8 | 4.6<br>4.8<br>5.0 | 4.6<br>4.8 | 4.6<br>4.8 | | | | |
| Dissipation factor | A<br>D-24/23<br>D-48/50 | 4.6.6 | Max. | 0.035<br>0.040 | 0.035<br>0.040 | 0.035<br>0.040 | 0.035<br>0.040<br>0.045 | 0.035<br>0.040 | 0.035<br>0.040 | | | | |
| Volume resistivity | C-96/35/90 | 4.6.7 | Min.<br>megohm<br>cm | | | | | | | | | | |
| Surface resistance | C-96/35/90 | 4.6.8 | Min.<br>megohm | | | | | | | | | | |
| Arc resistance | D-48/50 | 4.6.9 | Min.<br>sec. | | | | | | | | | | |
| Tracking resistance | A | 4.6.10 | Min.<br>minutes | | | | | | | | | | |
| Impact strength<br>Lengthwise:<br>Crosswise: | E-48.50 | 4.6.11 | Min.<br>ft-lbs.<br>per inch | | | | | | | | | | |
| Flexural strength<br>Lengthwise:<br>Crosswise: | A | 4.6.12 | Min.<br>psi | 20,000<br>16,000 | 20,000<br>16,000 | 20,000<br>16,000 | 20,000<br>16,000 | 20,000<br>16,000 | 20,000<br>16,000 | | | | |

See footnotes at end of table.





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TABLE II. *Performance requirements for sheets – Continued.*

| Requirement | Cond. | Test para. | Unit | Thickness (inches)[1] | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 0.031 | 0.062 | 0.094 | 0.125 | 0.188 | 0.250 | 0.500 | 0.750 | 1.000 | 1.001-Max. |
| Thermal endurance Flexural strength: | | 4.6.13 | Min. psi | | | | | | | | | | |
| Bonding strength[2] | A D-48/50 | 4.6.15 | Min. pounds | | | | | | | | | | |
| Water absorption | $D_1$-24/23 | 4.6.17 | Max. percent | 1.00 | 0.65 | 0.60 | 0.50 | 0.40 | 0.25 | | | | |
| Silicone content | E-168/2685 | 4.6.18 | Max. ppm | | | | | | | | | | |
| Flame resistance | | | | | | | | | | | | | |
| Ignition time: | A | 4.6.19 | Min. sec. | | | | | | | | | | |
| Burning time: | | | Max. sec. | | | | | | | | | | |

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.





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Custodians:
Army – ER
Navy – SH
Air Force – 20

Review Activities:
Army – MI, AR, EA
Navy – EC, OS
Air Force – 85

User Activities:
Army – ME
Navy – MC, AS

Preparing Activity:
Navy – SH
(Project 5970-1015-26)



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